# TIM-LOUIS WILKEN

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## EDUCATION

**UNIVERSITY COLLEGE LONDON (UCL) / UNIVERSITY OF OXFORD** — London / Oxford, United Kingdom
*M.Sc. Energy Systems and Data Analytics (UCL); thesis as visiting student (Oxford)* — *Sep 2025 – Sep 2026*

- **Average 76.8/100** across all taught modules (Distinction $\geq 70$; dissertation pending). Top marks: Statistics for Energy Analytics **89**, Energy and Transport Analytics **85**, Econometrics of Energy Markets **82**.
- Thesis, written at Oxford (Jun–Sep 2026, fee waiver), supervised by A. Stratigakos (UCL), T. Morstyn and Y. Zhou (Oxford): *Deadline Commitments for GPU Jobs under a Forecast-Uncertain Non-Firm Grid Connection*. Two-stage stochastic MILP with ECMWF ensembles and chance constraints; 6–21 pp more deadlines than a greedy scheduler.

**TECHNICAL UNIVERSITY OF MUNICH (TUM)** — Munich, Germany
*B.Sc. Management and Technology* — *Oct 2021 – Mar 2025*

- Final grade 2.2 (German scale, 1.0 = best); electrical engineering, business analytics and finance; exchange semester at Singapore Management University (2023).
- Thesis (grade 1.3), *Rising connection requests for BESS in Germany*: analysed 2024 German day-ahead price volatility and battery cost degression behind 340 GW of BESS grid-connection requests, quantifying arbitrage revenues.

**SIMON-MARIUS-GYMNASIUM** — Gunzenhausen, Germany
*Abitur; DPG-Abiturpreis (German Physical Society) for outstanding achievement in physics* — *2012 – 2020*

## TECHNICAL PROJECTS

- **GB Day-Ahead Price Forecasting**: online ensemble of 13 models (LSTM, Transformer, gradient boosting, LEAR and others), retrained daily and combined via Bernstein Online Aggregation; 21 % lower MAE than the LEAR benchmark over 2024–2026 (Diebold–Mariano $p < 0.001$); calibrated probabilistic forecasts; in a day-ahead battery backtest (50 MW / 100 MWh) it captures 93–94 % of the perfect-foresight arbitrage revenue, vs. 90 % for LEAR and 75 % for persistence; strict leakage audit (pre-auction data only). *(Python, PyTorch, scikit-learn; live daily at* timlouiswilken.github.io/gb-dayahead-forecast*)*
- **BESS Trading Simulator**: web app for GB battery trading across day-ahead, intraday and imbalance markets on live Elexon BMRS data, with balancing-mechanism acceptances and a P&L dashboard. *(React, TypeScript; built with a fellow student, 2026;* bess-trading-simulator.com*)*
- **Australian Electricity-Market Dynamics**: three-regime threshold vector error-correction (TVECM) analysis of how battery storage shapes SA–VIC wholesale price spillovers (52,608 half-hourly observations, 2SLS). *(Python, statsmodels;* github.com/TimLouisWilken/analysis-of-the-australian-electricity-market*)*

## PROFESSIONAL EXPERIENCE

**JOH. BERENBERG, GOSSLER & CO. KG** — Munich, Germany
*Origination and Growth Intern (full-time)* — *May 2025 – Aug 2025*

- Built data-driven lead-generation workflows across 5 client segments, scaling to ∼100 qualified leads per week.

**LUX-WERFT UND SCHIFFFAHRT GMBH** — Pleinfeld, Germany
*Ship Operator, MS Brombachsee, Europe's largest inland passenger trimaran (part-time)* — *Jun 2024 – Aug 2025*

**MOONROC ADVISORY PARTNERS GMBH** — Munich, Germany
*Strategy Consulting Intern (full-time)* — *Feb 2024 – May 2024*

- Quantified interview data for strategy workshops; co-authored *Retail Banking Compass 2024* (fintech chapter).

**KPMG AG (AUDITING & ADVISORY)** — Munich, Germany
*Research Assistant, Audit (full-time)* — *May 2023 – Aug 2023*

- Developed a client toolkit for sustainability reporting with TUM's Chair of Financial Accounting.

**WEBSHAPE WEBDESIGN** — Munich, Germany
*Co-Founder & Managing Director (self-employed)* — *Mar 2020 – Oct 2022*

- Co-founded a profitable web-design agency with high five-figure revenue, delivering 35+ client projects.

## AWARDS & LEADERSHIP

- **Rolls-Royce University Project** (2025–26): reactor systems design (HTGR); team awarded first place.
- **Cambridge Climate Challenge** (2026): team lead of a three-person interdisciplinary team; finalist.
- **UCL Energy Society** (2025–26): External Speaker Officer, sourcing industry experts for panels and keynotes.

## SKILLS

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| **Energy markets** | Day-ahead, intraday and imbalance markets (GB, DE); BESS arbitrage; Elexon and ECMWF data. |
| **Methods** | Probabilistic forecasting; stochastic & mixed-integer optimisation; time-series econometrics; ML. |
| **Tools** | Python (pandas, PyTorch, scikit-learn, statsmodels, PuLP); HiGHS, Gurobi; SQL, R, Git, TypeScript. |
| **Languages** | German (native), English (C1). |